

Returns, Volatility and Predictability - Can Volume Help?

Evidence from the Japanese Stock Market

by Patrick Natarajan Larsen

Student No. 119601

May 28, 2025

Abstract

In this paper, I analyze the returns and volatility of the Nikkei 225 stock index in the period 2016-01-01 to 2025-03-31. I find that the univariate mean equation follows a random walk specification with no drift but includes a COVID dummy. Diagnostics of the error terms reveal strong ARCH effects, and the most adequate model is the asymmetric TARCH model that captures the leverage effects of financial returns: negative shocks amplify volatility by a factor of 1.057, versus 0.104 for positive shocks. In an attempt to exploit additional data, I estimate a bivariate VAR(5) of returns and trading volume of the index. Although lagged volume is statistically significant in the VAR(5), the model explains less than 3% of the variation in returns, implying limited economic relevance. Out-of-sample, one-step-ahead rolling forecasts yield RMSE of 0.01354 for TARCH versus 0.01362 for VAR, with no significant difference per a Diebold–Mariano tests. Therefore, I conclude that asymmetric volatility modeling is essential for capturing the volatility dynamics of Japanese equities, while trading volume adds negligible predictive power, consistent with weak-form market efficiency.

Keywords: Returns, Nikkei, ARIMA, ARCH, GARCH, VAR

1 Introduction

Since Harry Markowitz’s seminal work on portfolio selection (Markowitz, 1952), the predictability and volatility of stock returns have been central topics in financial economics. The interest was amplified when French et al., 1987 applied the GARCH framework, initially introduced by Bollerslev, 1986, to model time-varying volatility in financial returns. Understanding return predictability is crucial for both institutional and retail investors seeking to generate alpha, while accurate volatility modeling is equally vital for investors and regulators aiming to assess portfolio risk through measures such as Value at Risk. In this paper, I aim to explore which types of volatility models adequately fit the dynamics of returns on the Japanese stock index. Furthermore, I investigate whether the returns are predictable, utilizing past realizations of returns and the trading volume of the index.

2 Literature Review

The classical approach to forecasting returns and their volatility is typically conducted in a univariate framework, most commonly using GARCH models or their extensions (see, for example, Akgiray, 1989; Pagan and Schwert, 1990). However, one might argue that modeling a series as complex as stock returns in a univariate setting is overly simplistic. If returns were truly predictable in such a framework, this would challenge Eugene Fama’s Efficient Market Hypothesis in its weak form (Fama, 1970).

A multivariate approach that has received significant attention is the relationship between trading volume and returns, as volume is often interpreted as a proxy for the rate of information arrival or investor sentiment. Unfortunately, empirical findings in this area remain ambiguous, often varying with market structure, geographic location, and asset size (Bajzik, 2021). R. M. Copeland, 1976 suggests a positive correlation between volume and price changes, driven by the sequential arrival of information. In contrast, Odean, 1998 examines behavioral explanations, highlighting how overconfident traders contribute to excess trading volume and increased return volatility.

3 Data

The data is obtained from Bloomberg and consists of daily observations of the *closing* price and *aggregated intraday* volume of the Japanese stock index, Nikkei 225, throughout the period 2016-01-01 to 2025-03-31. I follow Tsay, 2010, p. 5, and model the natural logarithm of the series, such that the first difference is equivalent to the continuously compounded return $r_t = \ln(P_t/P_{t-1}) = \ln(P_t) - \ln(P_{t-1})$. The *training* dataset consists of the period 2016-01-01 to 2022-12-31 corresponding to 75 percent of the entire dataset or 1730 observations, while

the test dataset corresponds to the period 2023-01-01 to 2025-03-31.

Among the relevant summary statistics in table 1 the skewness and kurtosis of the return series are most notable. The distribution exhibits negative skewness and excess kurtosis $k(x) = 6.78$, indicating fatter tails than the Gaussian distribution. Negative skewness and leptokurtosis are fairly prevalent features of financial returns and are both observations that motivated a wide range of extensions to the classical Black Scholes model. These distributional features are important considerations when selecting an appropriate volatility model.

Table 1: Summary statistics for Price, Volume, and Returns (Entire Dataset)

Variable	N	Mean	SD	Min	Max	Skew	Kurtosis
Price	2278	25,750.80	6,564.19	14,952.02	42,224.02	0.65	-0.44
Volume	2278	9.29×10^7	3.61×10^7	3.16×10^7	3.41×10^8	1.47	3.59
Return	2278	0.00	0.01	-0.13	0.10	-0.51	9.78

Confer figure 1, the training dataset contains a structural break in February 2020 around the COVID-19 pandemic, coinciding with the earliest reports of fatalities in Japan (World Health Organization, 2020). I model this with a prolonged pulse dummy, following the definition of Enders, 2014, pp. 263-264. The length of the period does require some attention, and two candidates are relevant. A short dummy, capturing the initial decline of the stock market until its bottom thus $D_1 = 1$ when $2020-02-20 < t < 2020-03-23$. Alternatively, a long dummy capturing the period of decline until recovery: $D_2 = 1$ when $2020-02-20 < t < 2020-06-19$. I shall use the former, as the latter may suffer from overfitting normal market dynamics of a rebound effect. Furthermore, a dummy version of the Chow test finds the long dummy insignificant for the returns series, which is the primary variable of interest for this paper.

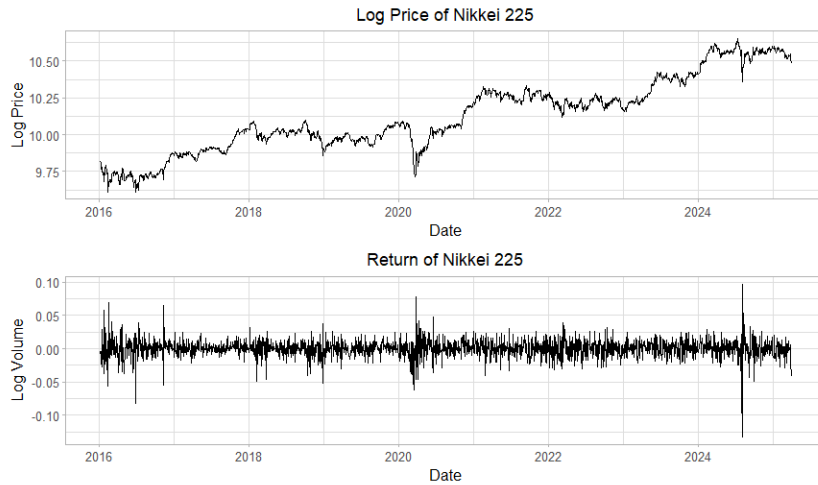


Figure 1: Price and Return Development of the Nikkei 225

4 Methodology

4.1 Stationarity & the Mean Equation

In order to apply the Box Jenkins Methodology for modeling the mean equation of a stochastic variable y_t we require that it is *covariance stationary*, meaning that the mean and all autocovariances are constant such that $\forall t, t-s: \mathbb{E}(y_t) = \mathbb{E}(y_{t-s}) = \mu, \text{Var}(y_t) = \text{Var}(y_{t-s}) = \sigma_y^2, \text{Cov}(y_t, y_{t-s}) = \gamma_s$. A sufficient condition for covariance stationarity of an ARMA(p,q) process:

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \dots a_p y_{t-p} + \epsilon_t + b_1 \epsilon_{t-1} + b_2 \epsilon_{t-2} + \dots b_q \epsilon_{t-q}$$

Or in lag operator form:

$$\underbrace{(1 - a_1 L - a_2 L^2 - \dots a_p L^p)}_{\text{Characteristic Polynomial}} y_t = a_0 + (1 - b_1 L - b_2 L^2 - \dots b_q L^q) \epsilon_t$$

is that the roots of the characteristic equation lie *inside* the unit circle, or equivalently that the roots of the characteristic polynomial lie *outside* the unit circle (Hamilton, 1994).

4.2 Random Walk

Although the operational definition of stationarity does not limit itself to the study of roots lying *on* the unit circle, this is often the case in the financial literature as time series often follow a so called random walk:

$$y_t = y_{t-1} + \epsilon_t$$

Modeling y_t as is, will prove to be an issue as the error terms persist, which is easily seen by solving the equation with backwards induction and an initial condition y_0 :

$$y_t = y_{t-2} + \epsilon_{t-1} + \epsilon_t = \dots = y_0 + \sum_{i=1}^t \epsilon_i$$

Often, alternative specifications that include a drift or trend component are also seen, such that y_t follows:

$$y_t = a_0 + y_{t-1} + \epsilon_t$$

$$y_t = a_0 + y_{t-1} + a_1 t + \epsilon_t$$

4.3 Augmented Dickey Fuller

With the prevalence of roots on the unit circle, it is straight forward to consider testing for such relationship by testing for the null hypothesis $H_0 : a_1 = 1$ in: $y_t = a_1 y_{t-1} + \epsilon_t$ or similarly whether $\pi_1 = 0$ in $\Delta y_t = \pi_1 y_{t-1} + \epsilon_t$ where $\pi_1 = a_1 - 1$. which would correspond

to the classic Dickey Fuller test. This test however, does not allow for other forms (such as the drift and trend equations above), nor for autocorrelation in the error term. Thus, one must mend the residuals and allow for trend and drift components in the mean equation by utilizing an Augmented Dickey Fuller test. That is, by running the regression:

$$\Delta y_t = a_0 + \gamma y_{t-1} + \beta_2 \Delta y_{t-2} + \dots \beta_p \Delta y_{t-p-1} + \epsilon_t$$

and testing for the null of a unit root by $H_0 : \gamma = 0$.

4.4 ARCH Effects

A standard assumption in classical econometric modeling is that the error term exhibits constant variance, known as homoskedasticity. For many econometric applications however, volatility moves in clusters and exhibits intertemporal dependence, despite its lack of serial correlation. That is, the series shows periods of unusually large volatility and periods of relative tranquility (Enders, 2014, p. 123). As in the cross-sectional space, a violation of homoskedasticity produces *consistent* albeit *inefficient* estimators. ARCH models capture this behavior by modeling the mean and variance of a series simultaneously. The conditional variance is allowed to be auto-regressive, such that it utilizes recent past realizations of the errors. To be more specific, in the class of multiplicative conditionally heteroskedastic models, $y_t = \mu + u_t$ where $u_t = \sigma_t \epsilon_t$ and $\epsilon_t \sim \text{iid}$ we allow σ_t to follow an auto-regressive process:

$$\sigma_t^2 = a_0 + a_1 u_{t-1}^2 + \dots a_p u_{t-p}^2$$

such that the conditional variance: $\mathbb{E}(u_{t+1}^2 | \mathcal{I}_t) = a_0 + a_1 u_{t-1}^2 + \dots a_p u_{t-p}^2$. where $\mathcal{I}_t$ refers to the information set at time t . The ARCH class of models has a handful of weaknesses, two of which we shall address. Firstly, the appropriate order of the ARCH models are often high, which leads to a lack of parsimony in the finalized model. Furthermore, as the errors are squared, negative and positive shocks are assumed to have the same effect on volatility (Tsay, 2010, p. 119, 131).

4.5 GARCH Effects and their Derivatives

In order to address the issue of parsimony, I shall follow Bollerslev, 1986 who defined the Generalized ARCH process by allowing for the conditional variance to follow an ARMA process. Specifically, consider again the multiplicative model $u_t = \sigma_t \epsilon_t$, but let σ_t follow a GARCH(p,q) structure:

$$\sigma_t^2 = a_0 + a_1 u_{t-1}^2 + \dots + a_p u_{t-p}^2 + b_1 \sigma_{t-1}^2 + \dots + b_q \sigma_{t-q}^2$$

As the specific order of the GARCH model is difficult to determine, I follow Tsay, 2010, p. 134 and iterate over low order models to determine the most appropriate specification.

While GARCH models offer a more parsimonious alternative to high-order ARCH models, they still assume that positive and negative shocks to u_t have the same impact on conditional volatility, due to the squaring of residuals in the variance equation. This is often violated in practice, especially for financial returns, as negative shocks tend to increase the variance more than positive ones. One can test for the presence of such leverage effect following (Enders, 2014, p. 157). In order to allow for the presence of such effects, I implement a threshold ARCH model (TARCH) with threshold $u_{t-1} = 0$. That is, the dummy $d_{t-1} = 1$ if the past shock is less than the threshold in the equation:

$$\sigma_t = a_0 + a_1 u_{t-1}^2 + d_1 \lambda_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}$$

such that a positive shock has impact a_1 while a negative shock has impact $a_1 + \lambda_1$.

4.6 Vector Autoregressive Models

One approach to modeling the relationship between variables in a multivariate setting, concerns fitting the relationship of, say a transfer function of the form $y_t = A(L)y_{t-1} + C(L)x_t + B(L)\epsilon_t$ where x_t is some exogenous variable of interest. If the relationship between x_t and y_t is endogenous, say, through feedback or reverse causality mechanisms, such estimators will necessarily be biased and inconsistent. Instead, consider the first order *structural* VAR(1) (Enders, 2014):

$$y_t = b_{10} - b_{12}x_t + \gamma_{11}y_{t-1} + \gamma_{12}x_{t-1} + \epsilon_{yt}$$

$$x_t = b_{20} - b_{21}y_t + \gamma_{21}y_{t-1} + \gamma_{22}x_{t-1} + \epsilon_{xt}$$

Which we can transform into a reduced form VAR(1):

$$\begin{bmatrix} 1 & b_{12} \\ b_{21} & 1 \end{bmatrix} \begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} b_{10} \\ b_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{yt} \\ \epsilon_{xt} \end{bmatrix}$$

$$BY_t = \Gamma_0 + \Gamma_1 Y_{t-1} + \epsilon_t$$

$$Y_t = B^{-1}\Gamma_0 + B^{-1}\Gamma_1 Y_{t-1} + B^{-1}\epsilon_t = \Phi_0 + \Phi_1 Y_{t-1} + u_t$$

Estimating the reduced form VAR is possible through OLS or MLE, but identification of the contemporaneous parameters in a structural VAR model necessitates imposing restrictions on a parameter. A common approach, is to assume a recursive structure (Cholesky Decomposition) by restricting $b_{12} = 0$, such that B is lower triangular. Assuming no use of the so called *Near VAR* approach, the h step ahead forecast is directly available: $\mathbb{E}(Y_{t+h}|I_T) = \Phi_0 + \Phi_1 Y_{T+h-1}$ and as in the univariate case, impulse response functions can be computed directly from the moving average representation of Y_t (Enders, 2014, pp. 291–295)

5 Results

5.1 Addressing Stationarity

As seen in Figure 1, the price level indicates trending behavior, violating the stationarity assumptions described in Section 4.1. Furthermore, when the series includes a structural break, the ADF test is biased towards accepting the null of a unit root. However, when the break date is known, the sample is sufficiently large, and the break spans more than a single period, it is valid to test for unit roots within each sub-sample separately (Enders, 2014, p. 227–229).

Table 2: Augmented Dickey Fuller Test Results for Price and Returns

Series/Test	Full Sample		Subsample 1		Subsample 2	
	Statistic	5% Critical	Statistic	5% Critical	Statistic	5% Critical
Price						
ADF with drift	-1.60	-2.86	-1.37	-2.86	-2.75	-2.86
ADF with trend	-3.07	-3.41	-2.73	-3.41	-2.04	-3.41
ADF with none	0.67	-1.95	0.70	-1.95	0.95	-1.95
KPSS (μ)	15.16 ^{***}	0.463	9.52 ^{***}	0.463	4.41 ^{***}	0.463
Returns						
ADF with drift	-28.22 ^{***}	-2.86	-23.14 ^{***}	-2.86	-19.25 ^{***}	-2.86
ADF with trend	-28.21 ^{***}	-3.41	-23.13 ^{***}	-3.41	-19.35 ^{***}	-3.41
ADF with none	-28.21 ^{***}	-1.95	-23.13 ^{***}	-1.95	-19.24 ^{***}	-1.95
KPSS (μ)	0.04	0.463	0.05	0.463	0.60	0.463

For the price level, the ACF exhibits strong persistence in both sub-samples and for the full sample while the return series shows exponential decay in the ACF and no significant spikes in the PACF. This indicates that the returns are stationary while the price level is not. Table 2 reports the appropriate ADF tests with drift and none for the two series, as well as alternative specifications and a KPSS test to verify the robustness of the results. The null hypothesis of a unit root in the return series is rejected for all specifications of the ADF test.

5.2 Random Walk and Conditional Heteroskedasticity

As there are no significant spikes in the ACF and PACF of the return series, the Box-Jenkins methodology suggests a random walk specification $\ln(P_t) = \ln(P_{t-1}) + \epsilon_t$. The `auto.arima` function in R (Hyndman and Khandakar, 2008) finds a similar result, a random walk with no drift:

$$\ln(P_t) = \ln(P_{t-1}) + -0.0364\text{Covid} + u_t \quad (1)$$

A Ljungbox Q Test statistic $Q = T(T+2) \sum_{k=1}^s r_k^s / (T-k)$ does not exceed the critical value in an appropriate χ^2 distribution for lags 10 through 50. Thus, we cannot reject the null of

zero remaining autocorrelation.

As addressed in section 4.4, heteroskedastic variance often manifests itself as an autoregressive process in the squared error terms. Presence of such effects, will make the above model inefficient.

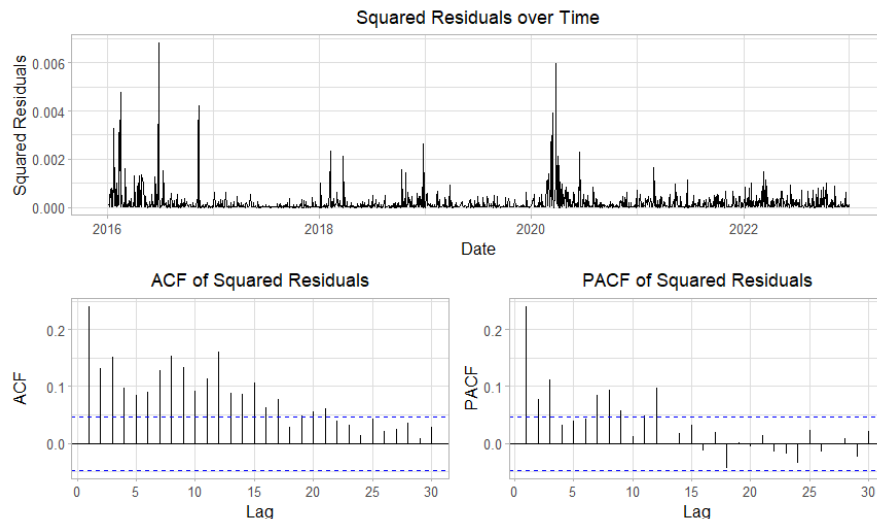


Figure 2: Evidence of ARCH Effects in Squared Residuals

As seen in figure 2 the squared residuals show periods of high volatility and relative tranquil and their PACF has significant spikes indicating intertemporal dependence despite the lack of serial correlation. The spikes are significant at lag 1, 2, 3, 7 and 8, meaning these models will be tested. Following Tsay, 2010, p. 134 the ARCH models are challenged with parsimonious GARCH models as well as a TARCH model to address the leverage effects discussed in section 4.5.

Table 3: Summary of GARCH Model Fit Results

Model	AIC	BIC	Ljung-Box (Lag[1])	ARCH LM Test	Sign Bias (Joint)
ARCH(1)	-6.0014	-5.9920	0.9996 (0.3174)	16.49 (4.88e-05)	4.0565 (0.2554)
ARCH(2)	-3.1992	-3.1866	0.0006 (0.9811)	0.0006 (0.9808)	2.7620 (0.4298)
ARCH(3)	-4.2709	-4.2552	0.0002 (0.9894)	0.0006 (0.9808)	0.9034 (0.8246)
ARCH(7)	-6.0748	-6.0464	0.4917 (0.4832)	3.794 (0.0514)	5.9860 (0.1123)
ARCH(8)	-6.0853	-6.0538	0.4072 (0.5234)	1.960 (0.1615)	5.4541 (0.1414)
GARCH(1,1)	-6.0966	-6.0840	0.5650 (0.4523)	0.1774 (0.6736)	8.4548 (0.0375)**
GARCH(1,2)	-6.0962	-6.0804	0.5834 (0.4450)	0.7413 (0.3892)	7.9163 (0.0478)**
GARCH(2,1)	-6.0958	-6.0800	0.5673 (0.4513)	0.9537 (0.3288)	8.4278 (0.0379)**
GARCH(2,2)	-6.0950	-6.0761	0.5832 (0.4451)	2.567 (0.1091)	7.9150 (0.0478)**
TARCH(1,1)	-6.2091	-6.1902	0.1336 (0.7148)	0.06189 (0.8035)	1.6232 (0.6541)

The TARCH model has the best fit, as measured by its Akaike and Bayesian information criteria. The weighted Ljung-Box tests on the squared residuals, as well as the ARCH LM tests,

fail to reject the null hypothesis of no remaining autocorrelation or ARCH effects, indicating that the heteroskedastic variance has been adequately modeled. Notably, the GARCH(1,1) does the same, but exhibits a sign bias, indicating the presence of a leverage effect. Thus, the most appropriate model is the mean equation 1, where $u_t = \sigma_t \epsilon_t$, $\epsilon_t \sim \text{wn}$ and the following equation, where I allow non-normal error terms cf. the discussion in 3 and Enders, 2014, pp. 157-158:

$$\sigma_t = 0.000617 + 0.104102u_{t-1}^2 + d_{t-1}0.953133u_{t-1}^2 + 0.873663\sigma_{t-1} \quad (2)$$

Thus, positive shocks have an impact of $\alpha_1 = 0.104102$ while a negative shock has impact $\alpha_1 + \lambda_1 = 1.057235$ which adequately captures the expected leverage effect. The magnitude of λ_1 raises concerns about potential non-stationarity, which does warrant further investigation. However, given that this study focuses on short-term forecasts, I consider it beyond the scope of this paper to implement regime-switching or integrated GARCH models.

5.3 A Multivariate Approach

As I intend to utilize the volume of the Nikkei Index, an immediate concern is whether the relationship is exogenous or endogenous. Intuitively, the relationship is likely to exhibit a feedback mechanism: as returns increase, volume increases as investors take profits or enter the market. As volume increases, the price increases even further as new investors enter the market. The volume series appears to exhibit a unit root, both from visual inspection and an ADF tests. This is a concern, as modeling variables of different orders of integration is inappropriate. However, differencing a trend stationary process, will introduce a non-invertible unit root process into the MA component of the model (Enders, 2014, p. 192). I follow the work done by Gallant et al., 1992 and Lee and Rui, 2002, who find that volume is trend stationary. I test for this by running the regression $\ln(\text{volume}) = \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t$. All variables are statistically significant at the one percent level and the detrended volume variable is stationary cf. an ADF and KPSS test. I choose to include five lags in the VAR Model, as it corresponds to a full trading week. This is the prevalent approach in the literature (see for example Lee and Rui, 2002) and is also chosen by the varSelect command in R, at least for the AIC specification (Pfaff, 2008). The estimation yields the following equation, in which insignificant estimates are suppressed:

$$\begin{aligned} \log\text{return}_t = & -0.0414^* \log\text{return}_{t-1} + 0.00402^{**} \log\text{volume}_{t-1} \\ & - 0.0492^{**} \log\text{return}_{t-3} - 0.0404^* \log\text{return}_{t-5} \\ & + 0.000520^* \text{Covid} + \epsilon_t \end{aligned} \quad (3)$$

All roots of the characteristic polynomial lie inside the unit circle, indicating stationarity. Although lagged volume for the first lag is significant, the amount of total variation explained by

the model is negligible, with an adjusted R^2 of 0.029. Unfortunately, this is in line with Bajzik, 2021 who find that “*the effects of trading volume on the stocks of the banks or overall indexes turn to be insignificant*”. However, a Granger causality test finds that volume Granger-causes returns, suggesting that volume may hold predictive power for future returns. Although volume Granger-causes returns in a statistical sense, the effect does seem to be economically negligible.

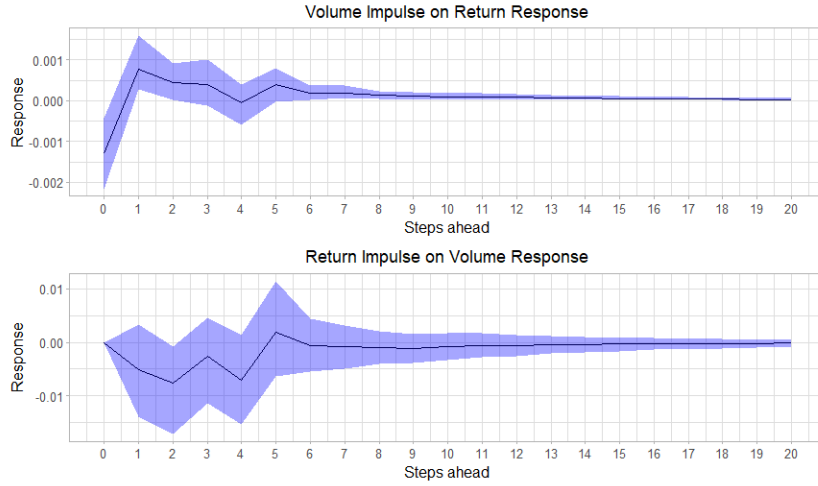


Figure 3: Impulse Response Functions of VAR(5)

Impulse response functions are computed, with bootstrapped confidence intervals and orthogonal error terms via a Cholesky decomposition. In this decomposition the contemporaneous effect of return on volume is restricted. Interestingly, figure 3 suggests that the contemporaneous effect of volume on return is negative before turning positive in the subsequent lag. This is noteworthy, as T. Copeland, 1976 suggests that sequential information arrival should make the contemporaneous relationship positive. As the effect is positive in the first lag however, it may also simply indicate, that information arrival is slower than anticipated. The multivariate Portmanteau test for remaining serial correlation is not rejected, indicating no remaining autocorrelation in the model. However, the Jarque-Bera and ARCH LM tests indicate rejection of normality and the presence of ARCH effects. This result is consistent with previous discussions regarding the distributional properties of financial returns. Although this does warrant further investigation, I consider it beyond the scope of this paper.

5.4 Forecasting

As the goal of the paper is to model the returns in the *short term*, I compute rolling one step ahead forecasts with recursive estimation of the models. This approach is the most reasonable from a practical point of view, and enables a visualization of the strength of the GARCH models. By utilizing the auto-regressive features of the variance, the model is able to capture the moments of relative tranquil and volatility as described in 4.4. This is visually captured by

the confidence bands shrinking or expanding during periods of tranquil and volatility respectively. Given that the estimated mean equation for the TAR(1,1) model is a random walk, I do not separately include a random walk benchmark, as it is effectively embedded in the model specification

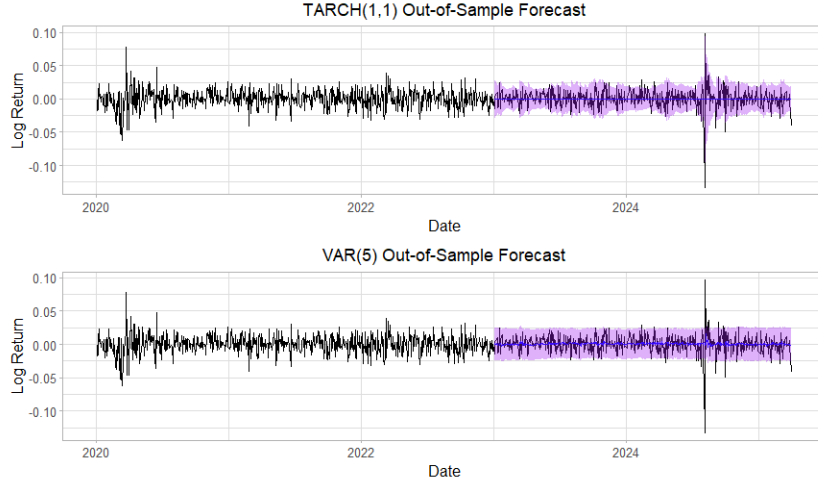


Figure 4: Rolling One-Step-Ahead, Out of Sample Forecasts

From the lower panel of figure 4, it is evident that the VAR model does not adequately capture the variation in the return series. This was to be expected, considering the low explanatory power of the return equation in section 5.3. To compare the models performance, I compute the mean squared error $MSE = \frac{1}{n} \sum_{i=1}^n e_i^2$ and the root mean squared error $RMSE = \sqrt{MSE}$. The TAR(1,1) model performs marginally better for both metrics $RMSE_{TAR(1,1)} = 0.0135432$, $RMSE_{VAR(5)} = 0.01362451$. Finally, I compute a Diebold Mariano test, to verify whether the forecasts are significantly different from each other. Utilizing a quadratic loss function $d_i = g(e_{1i}) - g(e_{2i})$ where $g(e_i) = e_i^2$. The intercept in the regression $d_i = a$ is insignificant, even when utilizing robust standard errors. Thus, we cannot reject that there is no difference in the forecasting performance of the two models.

6 Conclusion

This paper seeks to investigate the predictability of the return of the Nikkei 225, as well as its volatility dynamics and relationship with trading volume. I find that the returns exhibit significant heteroskedasticity, and that one must utilize an asymmetric GARCH model with student-t errors to adequately capture the leverage effects, leptokurtic distribution and negative skewness of the returns series. The VAR model of returns and volume does seem to capture a statistically significant relationship between the two variables, however the effect is economically negligible, as evidenced by the model's low explanatory power and ability to improve the out-of-sample forecasts.

References

- Akgiray, V. (1989). Conditional heteroscedasticity in time series of stock returns: Evidence and forecasts. *The Journal of business (Chicago, Ill.)*, 62(1), 55–80.
- Bajzik, J. (2021). Trading volume and stock returns: A meta-analysis. *International review of financial analysis*, 78, 101923–.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. [https://doi.org/https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/https://doi.org/10.1016/0304-4076(86)90063-1)
- Copeland, R. M. (1976). A model of asset trading under the assumption of sequential information arrival. *The Journal of finance (New York)*, 31(4), 1149–1168.
- Copeland, T. (1976). A model of asset trading under the assumption of sequential information arrival. *The Journal of Finance*, 31(4), 1149–1168. Retrieved May 16, 2025, from <http://www.jstor.org/stable/2326280>
- Enders, W. (2014). *Applied econometric analysis*. Wiley.
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383–417. Retrieved May 17, 2025, from <http://www.jstor.org/stable/2325486>
- French, K. R., Schwert, G., & Stambaugh, R. F. (1987). Expected stock returns and volatility. *Journal of Financial Economics*, 19(1), 3–29. [https://doi.org/https://doi.org/10.1016/0304-405X\(87\)90026-2](https://doi.org/https://doi.org/10.1016/0304-405X(87)90026-2)
- Gallant, A. R., Rossi, P. E., & Tauchen, G. (1992). Stock prices and volume. *The Review of Financial Studies*, 5(2), 199–242. Retrieved May 16, 2025, from <http://www.jstor.org/stable/2962030>
- Hamilton, J. D. (1994). *Time series analysis*. Princeton University Press.
- Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for R. *Journal of Statistical Software*, 27(3), 1–22. <https://doi.org/10.18637/jss.v027.i03>
- Lee, B.-S., & Rui, O. M. (2002). The dynamic relationship between stock returns and trading volume: Domestic and cross-country evidence. *Journal of banking & finance*, 26(1), 51–78.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. Retrieved May 2, 2025, from <http://www.jstor.org/stable/2975974>
- Odean, T. (1998). Volume, volatility, price, and profit when all traders are above average. *The Journal of finance (New York)*, 53(6), 1887–1934.
- Pagan, A. R., & Schwert, G. (1990). Alternative models for conditional stock volatility. *Journal of econometrics*, 45(1), 267–290.
- Pfaff, B. (2008). Var, svar and svec models: Implementation within R package vars. *Journal of Statistical Software*, 27(4). <https://www.jstatsoft.org/v27/i04/>

Tsay, R. S. (2010). *Analysis of financial time series*. John Wiley & Sons, Inc. <https://doi.org/10.1002/9780470644560>

World Health Organization. (2020, February). Situation Report 31 [Retrieved 20 February 2020].